

Student ID: _____ Student Name: _____

Instructions: Usage of unfair means (UFM), including Mobile and Internet, is not allowed in the exam.
No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. Q1-20 are of 0.5 marks each. Max Duration: 15 minutes.

1. Arrange these correctly when the **system boots**: BIOS, Power On, Boot Loader, OS image.

Power On, BIOS, Boot Loader, OS Image

2. What are the **maximum permissions** associated with a file and directory using r,w,x notation?

rw- for file (6) rwx for directory (7)

3. Which Linux directory contains **per-process information** such as virtual memory, stack, limits, etc?

/proc

4. Type the command to get the process ID of the **current shell**.

echo \$\$

5. Within **Vi** editor, mention the command to **replace all occurrences** of Average with Good

:%s / Average / Good / g

6. How do you **locate multiple binaries** of the same name canary in Linux using shell/terminal?

locate canary

7. How do you help Rahul to **list all shells** present in the system?

cat /etc/shells

8. Ankit is struggling to **recall a command**. However, he remembers it starts with ls. How do you locate such commands?

compgen -c ls

9. On Ubuntu, Suprio cannot recall if set the root password. How do you confirm if no **root password is set**?

sudo -i and then passwd -S

10. Nageshwar has a dual-booted system with Windows and Linux (Ubuntu). His system does not show the correct time as per the timezone set. He has booted into Ubuntu. What command should he use to **synchronize the clock**?

timedatectl set-local-rtc 1 --adjust-system-clock

11. How do you **find file systems** available on the Linux systems? Multiple file systems can co-exist.

df -hT or lsblk or fdisk -l or mount

12. Help Pranjal to **hide** a **command** typed **from bash history**. It should not be stored and returned in history.

prefix a space before command

13. Name two **package managers** used in Linux systems, along with the distribution name.

Ubuntu - apt, dpkg

14. How can Hardik **delete** the current and the following 10 **lines** in the Vi editor?

10dd in command mode

15. Ayan remembers the command name. However, he is unsure about some of the **command's options and arguments**. Give him two options to find what he needs on the Linux system.

man pages, info command

16. Ashish wants to see which process takes the **maximum CPU time and memory** at any instant in Linux. Suggest one option.

top

17. Devansh discovered that his **disk partition was almost full**. He recalls one large file in his home directory that he can delete, however rm is not working. Help him to get rid of the contents of /home/parth/game.tar

> /home/parth/game.tar

18. Vibhav wants to **share some files on the same network** using an HTTP server. He has Python installed. Help him assuming there is no network restriction.

python3 -m http.server

19. How do you help Meet to find out the **space APT Cache is taking** in human-readable size? Hint.

/var/cache/apt

du -hs /var/cache/apt

20. How do you help Harshad to list the **partitions** on the disk? Command please.

fdisk -l or df -hT

Student ID: _____ Student Name: _____

Instructions: Usage of unfair means (UFM), including Mobile and Internet, is not allowed in the exam. No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. All questions carry equal marks. Max Marks: 10

A. Write regular expressions (regex) for the following (1 to 5):

1. Match all lines that start with \$, followed by any single digit, followed by \$, followed by one or more non-whitespace characters

`\$[0-9]\$[^\s]+`

2. Match every possible IPv4 IP address (use metacharacters and groups)

`([0-9]{1,3}\.){3}[0-9]{1,3}`

3. Match every 9-character string (with letters, numbers, and symbols) that doesn't end in a "!" sign

`[^\s]{8}[^!]`

4. Match PAN, examples AABPS0971Z, BBCYJ9876A, PANNO1234T

`[A-Z]{5}[0-9]{4}[A-Z]`

5. Match IFSC, contains 4 characters for bank name in capital, followed by one zero, followed by six characters in capital or numbers.

`[A-Z]{4}0[A-Z0-9]{6}`

6. How can you search a noun or verb using grep and ignoring "ing" in it? Use grep.

`grep -v "ing" text || cat text | grep -v "ing"`

7. How can you search a log file for ERROR and print five lines before and five lines after the matching line?

`grep -C 5 "ERROR" log.txt`

8. Write a sed command to replace all Unix references with Linux in the file input.txt

`sed 's/Unix/Linux/g' input.txt`

9. Write an awk command that adds a header "System Diagnostics Report", prints prtdiag.txt, and adds a footer "End of the Report"

`awk 'BEGIN { print "System Diagnostics Report" } { print $0 } END { print "End of the Report" }' prtdiag.txt`

10. Write an awk command that prints columns 1 and 2 of /etc/passwd using the separator ":"

`awk -F ":" '{ print $1 $2 }' /etc/passwd`

Indian Institute of Technology Jammu
Linux System Administration and Virtualization (ID-2-04(MO))
Mid Semester Exam (Session 2024-25)

Student ID: _____ Student Name: _____

Instructions: Usage of unfair means (UFM), including Mobile and Internet, is not allowed in the exam.

No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. Q1-10 1 Marks each, Q11-45 2 Marks each.

1. Which one of the following is not a shell?
(A) /bin/bash (B) /bin/dash
(C) /bin/rbash (D) /bin/rdash [D]
2. On /bin/bash, what the command !! will do?
(A) List history of commands (B) Repeat the last command
(C) Print the shell process ID (D) Clear the history of commands [B]
3. The following command locates the binary, source, and manual page files for a command?
(A) plocate (B) whereis
(C) which (D) whatis [B]
4. What is the numeric value of permissions for this file?
`crw-rw-rw- 1 root root 1, 3 Feb 20 17:24 /dev/null`
(A) 666 (B) 766
(C) 777 (D) 444 [A]
5. A log file keeps on updating and growing. How can we view its content while it is still getting updated?
(A) head file (B) tail -f file
(C) tail -z file (D) cat -t file [B]
6. Which of the following is incorrect about cd?
(A) cd takes to /home directory (B) cd ~ takes to home directory
(C) cd . . takes to parent directory (D) cd - takes to previous directory [A]
7. How to get a list of all signals on the linux, name and number?
(A) kill -l (B) kill -a
(C) kill -9 -a (D) kill %1 [A]
8. Which one of the below is an LTS release of Ubuntu?
(A) 22.10 (B) 23.04
(C) 23.10 (D) 24.04 [D]
9. In the ls -lrt output -rwsr-xr-x what is **s** meant for?
(A) symbolic link (B) sticky bit
(C) shared object (D) shell script [B]
10. Execute last command in ubuntu terminal that began with com
(A) runlast com (B) !com
(C) ^com (D) :&com [B]

11. On Ubuntu, rps cannot recall if he set the root password. How do you confirm if no root password is set?

sudo -i

passwd -S ⇒ check NP for no password set

12. rps has a dual-booted system with Windows and Linux (Ubuntu). His system does not show the correct time as per the timezone set. He has booted into Ubuntu. What command should he use to synchronize the clock?

timedatectl set-ntp true OR timedatectl set-local-rtc 0

13. cat command can be used to merge two files file1 and file2. Write the command.

cat file1 >> file2

14. rps has a compressed file ~/Downloads/json.zip and he wants to its compressed size, uncompressed size, compression ratio, and name of uncomprepsed file. Help him with correct command.

zcat -l json.zip zipinfo json.zip
~~gzip -l~~
~~gunzip -l~~

15. rps wants to view line number 1260 to 1265 in the file filename.txt Help him with correct command. Hint: He recalls something from head and tail.

head -1265 filename.txt | tail +1260

OR head -1265 filename.txt | tail -6

16. rps wants to share some files on the same network using an HTTP server. He has Python installed. Help him with correct command to start the server.

python3 -m http.server

17. Using read command, how can we show a prompt to Enter Name and read name in a variable sname?

\$ READ_COMMAND_HERE

Enter Name: Roger Southwood

\$ echo \$sname

Roger Southwood

read -p "Enter Name:" sname

18. Complete if statement in this bash script to match the output from echo.

file1=\$1

file2=\$2

IF_STATEMENT_HERE

echo "\$file1" is newer than "\$file2"

fi

if [\$file1 -nt \$file2]; then

19. How can standard output and standard error can be directed to file log.txt ?

&> log.txt

20. How to output command output of lsb_release -a to the terminal and a file called release.txt also?

lsb_release -a | tee release.txt

21. On /bin/bash terminal what is output of \$ yes

continuous text y until interrupted, liveness

22. What is the the difference between these two ways of execution of bash script?

\$ bash -v script.sh

\$ bash -x script.sh

echo/display command before running then

echo command after processing it

<command>

expands variables

<executed output>

<execute and output>

<command with expansion>

23. How will you count number of processors on the linux system using cpuinfo?

cat /proc/cpuinfo | grep -i "processor" | wc -l

24. How can we see the process memory layout of process ID 54321 using proc interface?

cat /proc/54321/maps

25. What does the output of the command show? ldd /bin/ls

shared objects and libraries of ls

26. How can we list out all the processes that belong to the logged in user?

ps -aef | grep \$USER

27. For local/current session in terminal how can we modify PATH variable to search home directory first for any executable?

export PATH=\$HOME:\$PATH

28. How can we find out Audio Device details on the Linux system? Hint: command starts with ls

lspci -v

29. How will you grant read permission to the group users and other users?

chmod -go+r <file>

30. How can we find out multiple users that belong to a group called: adm

cat /etc/group | grep -i "adm"

31. Write down permissions in numeric representation from the partial output shown.

lrwxrwxr-x 775 drwxr-xr-x 755
 -r----- 400 -rw-r--r-- 644

32. Everything is file in linux. The output of ls -lrt, column 1 lists file type. Fill in the blanks:

Regular file: - Directory: d
 Symbolic links: l Character device: c

33. A website url <https://www.mysite.com/uploads/> shows all files uploaded by users and anyone can delete them using HTTP DELETE [mysite.com/uploads/file](https://www.mysite.com/uploads/file) URIs. What is the issue here with uploads directory and how to fix it?

uploads directory has write permission for others.
Remove/revoke write permission for others.

34. How can we query all executables that have sticky bit set? Hint: find

find / -perm 4000

35. User has actually forgot the command. He recalls it starts with ip. Show him all commands that start with ip

compgen -A command ip

36. How can you ssh to a remote virtual machine using pem file without username and password?

ssh -i file.pem user@ip

37. How will you schedule a bash script to sync attendance /home/rps/sync_attendance.sh every night at 11 pm, except Sunday?

00 23 * * 1,2,3,4,5,6 rps { cd && ./sync_attendance.sh }

38. Why system administrators prefer "su -" over a simple "su" command to substitute user?

39. Why and how a non-root user is able to execute a command owned by root?

40. Using nmap how first 2000 ports can be scanned for url iitjammu.ac.in? Write the same using nmap for verbose output and using 4 threads.

nmap -p 1-2000 iitjammu.ac.in
nmap -p 1-2000 -vvv -Ty iitjammu.ac.in

41. How can we trace path from root name servers to local or nearest authoritative DNS server?

dig +trace iitjammu.ac.in

42. How can we enumerate all subdomains belonging to a domain myexample.net ?

dnsenum -r myexample.net
dnsrecon

43. Suppose we register a new domain nuvo.ac.in to domain registrar (ERNET). What two resource records are added by domain registrar in top-level-domain nameserver ns1.registry.in? Support authoritative dns server is dns1.nuvo.ac.in having public IP 103.27.32.1.

44. Write a bash script to sum natural numbers from 1 to 10.

45. Write a bash script to print the day of the week given a number. Sunday is 0, Monday to Friday are 1 to 6.

Student ID: _____ Student Name: _____

Instructions: Usage of unfair means (UFM), including Mobile and Internet, is not allowed in the exam.

No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. Q1-10 are of 0.5 marks each. Q11-30 are of 1 marks each. Maximum marks are 25 marks. Maximum Duration is 3 hours.

1. Event logs of the Linux system are stored at:
A) /var/log B) /etc/gub.conf C) /proc/stat D) /usr/sbin [**A**]
2. The option to run an interactive command on the remote system without a session disruption is:
A) nohup B) Run without stack C) Run without /proc D) ssh [**A**]
3. Which file system can be used to change certain kernel parameters at runtime using the sysctl command?
A) ext4 B) sysfs C) btrfs D) procfs [**D**]
4. The file system for CD-ROM is:
A) squashfs B) ntfs C) isofs D) ovaifs [**C**]
5. What is the utility to retrieve possible options for autocompletion of commands, builtins, arguments, etc.?
A) complete B) compgen C) which D) locate [**B**]
6. What is the primary role of the "inode" in a UNIX-based file system like EXT4?
A) It stores file data directly
B) It stores file metadata, such as file permissions, ownership, and pointers to data blocks
C) It tracks deleted files in the system
D) It acts as the root directory for the file system [**B**]
7. What is the main function of the /etc/shadow file in UNIX/Linux?
A) It stores user passwords in an encrypted format
B) It stores information about user home directories
C) It contains system logs and crash dumps
D) It tracks the system's performance metrics [**A**]
8. Which partitioning scheme uses a GUID Partition Table GPT instead of an MBR?
A) Windows FAT16, FAT32 based partitions
B) Modern UEFI-based systems
C) Linux ext2, ext3 based partitions
D) Legacy BIOS systems [**B**]
9. Which is a characteristic of solid-state drives (SSDs) compared to traditional hard drives (HDDs)?
A) Slower read/write speeds
B) Less reliable storage
C) Uses spinning disks
D) No moving parts [**D**]

10. Analysis of Linux shell history (.bash_history) shows a command in important_file.txt. However, file recovery tools find no trace of this file in unallocated space. What is the most plausible explanation?
- A) The rm command on Linux always securely wipes data.
 - B) The file system did not support journaling.
 - C) The .bash_history file cannot be trusted.
 - D) The file was renamed before deletion.
- [C]

11. While listing hidden files, ls -al command also lists the current directory (.) and the parent directory (..). How can we disable showing these two directories in the output while dealing with hidden files? Say the name of directory is /home/rps/docs/

`ls -al /home/rps/docs`

12. What will this command show? `cat /proc/self/maps`

process memory layout of cat executing this command.
 /usr/bin/cat code, text, post, heap, shared libs, stack, etc.

13. How can we list out all files on the system for which the sticky bit (setUserId) is set?

`find / -type f -perm /4000`

or `-4000` ✓

14. Bash history:

- a. How can we avoid commands from getting appended to history?

add a space before a command

e.g. `$ _cat /proc/self/maps`

- b. How to search bash history for command containing "bhoot"?

`ctrl+r` (reverse string search) and type bhoot

15. Using sed, how can we add a prefix string "2025" at the beginning of each line in a file called paper.txt?

`sed -i 's/^/2025/' paper.txt`

16. You are given a file id.txt that contains students' IDs. Using sed, how can you add a suffix string "@iitjammu.ac.in" at the end of each line in this file?

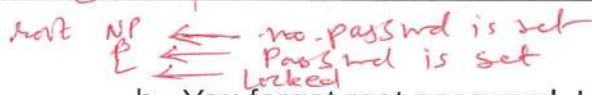
`sed -i 's/$/@iitjammu.ac.in/' id.txt`

17. How to get GPU information on a linux system? Mention two commands with options.

`lspci -v | grep -A7 -i "graphics"`

`lspci -v | grep "VGA"`

`lshw -C display`

18. Using secure copy, mention the command to
- transfer a file lab9.txt to a remote server 10.10.180.118 to home directory /home/rps of user rps
`scp lab9.txt rps@10.10.180.118:/home/rps/`
 - fetch a file game.txt from the remote server 10.10.190.128 that is residing in home directory /home/rambo of user rambo. The command should download this file to your home directory with name as endsem.txt
`scp rambo@10.10.190.128:/home/rambo/game.txt ~/endsem.txt`
19. You are tasked to test a website jksdm.pragyaware.com/ when you open this URL, the website is not opening, however developer says he is able to open the website and this is correct website. What can be the issue here and how to resolve this?
 Trying to open http website as https or ^{by} default url has https in it. Solution — open website as <http://jksdm.pragyaware.com/>
20. root user: or — install SSL/TLS certificate on server.
- How to check if root user password is set on Linux?
`$ sudo -i` on next prompt type `passwd -S`
`root@ — #passwd -S`

 - You forgot root password. How to delete it once you find out it was set?
 on root prompt `passwd -d root` to delete root password
21. Cron jobs: What these two lines in crontab exactly do?
- `0,10,20,30,40,50 * * * rps cd /home/rps/lav && touch a.out`
 Every 10th minute a.out file's timestamp is updated in /home/rps/lav
 - `00 7,13,19 * * * root cd /home/eg/backup && ./run_backup.sh`
 Every 7am, 1pm, 7pm run-backup.sh script is executed.
22. Write LaTeX command to produce the output given below:
 Baye's Theorem

$$P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

$\backslash \text{begin}\{\text{equation}\}$

$$\{ P(A/B) = \backslash frac\{ P(B/A) . P(A) \}\{ P(B) \} \}$$

$\backslash \text{end}\{\text{equation}\}$

Class Formula

$$\int_0^1 \tan(e^{x^{\sqrt{x} \log y}}) d\theta$$

$\backslash \text{begin}\{\text{equation}\}$

$$(\int_0^1 \tan(e^{x^{\sqrt{x} \log y}}) d\theta)$$

$\backslash \text{end}\{\text{equation}\}$

23. On a system Linux/Ubuntu is installed.
- How to open these files using the same COMMAND? We do not know what is the default application user has installed.

\$ COMMAND beamer.pdf
 \$ COMMAND calvinEducation.jpeg
 \$ COMMAND thankyou.png
 \$ COMMAND latex-package.zip
 \$ COMMAND https://saral.iitjammu.ac.in

$\downarrow$
 xdg-open

- How to send a desktop notification "This is LSAV endsem exam."

$\text{notify-send "This is LSAV endsem exam"}$

24. Write LaTeX commands to output the following table. Table number is automatic.

Table 1: Truth Table

m	t	TP / FN / FP / TN
T	T	TP
T	F	FN
F	T	FP
F	F	TN


```
\begin{table}[!htb]
```

```
\caption{Truth Table}
```

```
\centering
```

```
\begin{tabular}{|c|c|c|}\hline \hline
```

```
$\textbf{m}$ & $\textbf{t}$ & $\textbf{TP / FN / FP / TN}$ \\ \hline \hline
```

```
T & T & TP \\
```

```
T & F & FN \\
```

```
F & T & FP \\
```

```
F & F & TN \\ \hline
```

```
\hline
```

```
\end{tabular}
```

```
\end{table}
```

25. SSL/TLS Certificate:

- a. While setting up SSL/TLS certificate for Election Commission server ec.iitjammu.ac.in, which two files are required on the server?

certificate file .crt file

private key file .key file

- b. How will you check TLS version enabled and SSL/TLS certificate expiry?

sslsan ec.iitjammu.ac.in

↳ TLS version enabled

at bottom check dates for validity

26. File deletion and modification:

- a. System has accumulated journals and should be cleaned: journalctl shows 10 GB used up.

```
$ journalctl --disk-usage
```

Archived and active journals take up 10.0G in the file system.

How will you free up this space?

```
sudo journalctl --vacuum-time=1d
```


- b. Due to editing multiple files, and not using version control, you cannot figure out how many files were changed recently. How will you find out files modified in the last 24 hours?

find / -mtime 1

27. Curl:

- a. What will be output of this command?

```
curl -v -o /dev/null -s -w '%{time_total}s %{num_connects} \n'
https://saral.iiitammu.ac.in/
```

*Total time for request and response.
number of connections made*

- b. How will you check HTTP response headers for a URL using curl?

curl -v https://saral.iiitammu.ac.in

28. Convert and Merge:

- a. How to convert an image hate.png to PDF love.pdf

convert hate.png love.pdf

- b. How to merge three PDF files file1.pdf, file2.pdf and file3.pdf into one pdf file merged_file.pdf

pdftk file1.pdf file2.pdf file3.pdf cat output merged_file.pdf

29. A directory /home/rps/Downloads/ contains several WhatsApp Image files with jpeg extension, e.g.

'WhatsApp Image 2023-04-17 at 9.38.26 AM.jpeg'
'WhatsApp Image 2024-04-17 at 11.03.43 PM.jpeg'
'WhatsApp Image 2024-11-07 at 4.00.01 PM.jpeg'
'WhatsApp Image 2024-01-11 at 12.37.26 PM.jpeg'

Write a bash script to rename them to image-001.jpeg, image-002.jpeg, image-003.jpeg and so on in a new directory /home/rps/images

30. Write a bash script to create 10 users namely user1, user2, user3,...,user10, set their password as p@\$\$w0rd, this will be shared with them. Now, set the password to expiry so that they change the password upon login. Script should output users created, and password set to change upon login.

Indian Institute of Technology Jammu
Linux System Administration and Virtualization (ID-2-04(MO))
Lab End Semester Exam (Session 2024-25)

Student ID: _____ Student Name: _____

Instructions: Usage of unfair means (UFM), including Mobile and Internet, is not allowed in the exam.

No cheating. Any kind of cheating will get you a straight Fail grade in the course.

You are required to show output /execution for each task.

1. How do you find files modified on the system in the last two hours? Hint: 120 minutes.
2. While running an interactive command on the remote system, the session may get disconnected, or the terminal hangs up. How can the interactive command be ensured to be completed on the remote system and not end due to the disconnection of the terminal/session?
3. Show open ports on the given sites.
4. Show all sub-domains of iitjammu.ac.in domain.
5. Show the highest TLS version enabled on saral.iitjammu.ac.in and when the SSL certificate will expire.
6. Using your mypdf submissions, modify the section for file compression below a size to compress a file within a range of low and high values, i.e., Compress a file above 500 KB and less than 1 MB.
7. Create a two-page PDF document using LaTeX that should contain the following:
 - a. Title page with a heading and your name as author
 - b. Five Chapters and sub-sections. Use some dummy text from institute's website.
 - c. Two Images side-by-side with captions
 - d. Two Tables with captions
8. Install VirtualBox and download a Linux distribution of your choice. Create a VM with 2 GB of RAM, and allocate suitable resources to run this Linux distribution. After installation, login, and show the following:
 - a. Number of processors in it - commands only
 - b. RAM installed and available - commands only
 - c. File system - commands only
 - d. Transfer files from and to this VM and your local system
9. Mention one thing/topic you expected from this LSAV course that was not covered.

